

Micro-expression Generation with Thin-plate Spline Motion Model and Face Parsing

Jun Yu
University of Science and Technology
of China
Hefei, China
harryjun@ustc.edu.cn

Guochen Xie*
University of Science and Technology
of China
Hefei, China
xiegc@mail.ustc.edu.cn

Zhongpeng Cai
University of Science and Technology
of China
Hefei, China
czp_2402242823@mail.ustc.edu.cn

Peng He
University of Science and Technology
of China
Hefei, China
hp0618@mail.ustc.edu.cn

Fang Gao
Guangxi University
Guangxi, China
fgao@gxu.edu.cn

Qiang Ling
University of Science and Technology
of China
Hefei, China
qling@ustc.edu.cn

ABSTRACT

Micro-expression generation aims at transferring the expression from the driving videos to the source images, which can be viewed as a motion transfer task. Recently, several works have been proposed to tackle this problem and achieve great performance. However, due to the intrinsic complexity of the face motion and different attributes of face regions, the task still remains challenging. In this paper, we propose an end-to-end unsupervised motion transfer network to tackle this challenge. As the motion of the face is non-rigid, we adopt an effective and flexible thin-plate spline motion estimation method to estimate the optical flow of the face motion. What's more, we find that several faces with glasses show weird deformation in motion transferring. Thus, we introduce face parsing method to pay specific attention to the eyeglasses regions to ensure the reasonability of the deformation. We conduct several experiments on the provided datasets of the ACM MM 2022 micro-expression grand challenge (MEGC2022) and compare our method with several other typical methods. In comparison, our method shows the best performance. We (Team: USTC-IAT-United) also compare our method with other competitors' in MEGC2022, and the expert evaluation results show that our method performs best, which verifies the effectiveness of our method. Our code is available at <https://github.com/HowToNameMe/micro-expression>

CCS CONCEPTS

• Human-centered computing → Virtual reality; • Computing methodologies → Computer vision; Image segmentation.

*Corresponding author of this paper.

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KEYWORDS

Micro-expression, Motion Model, Thin-plate Spline, Face Parsing

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1 INTRODUCTION

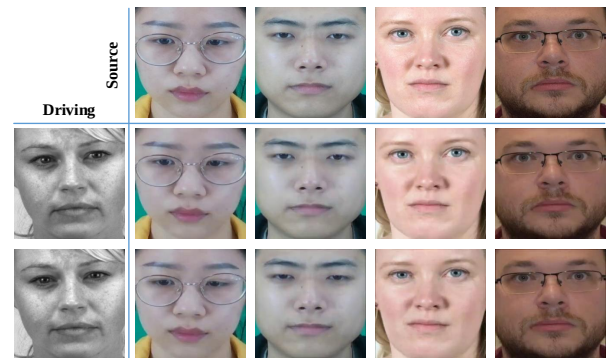


Figure 1: A brief introduction to the micro-expression generation task

Face expression has been a research focus for a long time for its wide application in multimedia analysis and human-computer interaction. In recent years, micro-expression generation task emerges and attracts the attention of many researchers. The micro-expression generation task is in the scenario of motion transfer task (see Fig. 1), which requires to transfer the motion from the driving video to a source image. These years, as the virtual reality (VR) technology develops, more and more VR based applications and products are designed and produced. Face expression generation, as one of the core tasks of the VR, are with broad prospect. Thus, the face expression task has been a research focus now.

As is mentioned above, the face expression generation task is subject to the motion transfer task. Therefore, most face expression generation methods are proposed by adopting or modifying

general motion transfer methods. Early motion transfer methods are mainly based on landmarks[2, 5, 6, 9] or action units[10]. These method used labeled data and generate images. However, due to the cost of the annotation, the usage of this method is limited. These years, unsupervised motion transfer methods emerge and gradually dominate this field. This type of method uses two frames sampled from the same video for training, where one frame is used as source image and the other frame is viewed as driving frame. The target of training is to reconstruct driving frame using source frame. And in the reconstruction process, the model can learn the motion representations.

MonkeyNet[11] is the typical models in this type. In MonkeyNet, one detector is adopted to learn several keypoints and then the difference between the point coordinates are used to estimate the optical flow of the motion. After that, the source image is fed into a generator to reconstruct the driving frame under the guidance of the optical flow. The method proposed a general framework for the latter researchers and its performance is also appealing. However, this model doesn't consider the missing region of the warped feature maps. To tackle this problem, FOMM[12] introduces additional occlusion maps. And the occlusion maps helps to focus on the inpainting of the missing regions. What's more, the FOMM method introduces a first order model to estimate the motion, which is more accurate compared to the previous zero order models (i.e. MonkeyNet).

These methods all achieve great performance. However, due to the intrinsic complexity of the facial motion, the representation ability of these linear models are not so adequate. Thus, inspired by the general motion transfer model[15], we attempt to adopt the nonlinear model to estimate the motion between source and driving images. More specifically, we first detect several keypoints of the face with the detector. and then, we conduct thin-plate spline (TPS)[1] estimation to approximate the motion from the source image to the driving image. Compared to the previous methods with linear estimation models, our method can model more complex deformation with the TPS estimation. After that, the source images are processed in the inpainting network to reconstruct the driving images under the guidance of the optical flow and occlusion maps. To make the generation process more accurate, we adopt the multi-scale architecture to warp and filter the feature maps in multiple layers.

Moreover, considering the different attributes of face parts, we need to pay specific attention to special regions to make sure that the deformation of this parts is reasonable and consistent to the global deformation. Among all the facial parts, the most distinct region of the face is the eyeglasses region, because the deformation of eyeglasses is rigid, while the deformation of all other facial parts are all non-rigid. Thus, in motion transfer process, the region of the eyeglasses is easily affected by the whole motion of the face and appear in a weird way. So, to gain more improvements in generation, we introduce the face parsing method to segment the region from the whole face, and then propose extra regression loss for this region to mitigate the irrational deformation of this region. We conduct experiments on the datasets of ACM MM 2022 micro-expression grand challenge (MEGC2022), and final results reveal that our method can generate satisfactory results with target

motion and outperform other methods. To summarize, our main contributions are as follows:

- We adopt the TPS estimation to model the facial deformation. Compared to the previous linear methods, our method is more effective in modeling the motion between the source and driving frames.
- As the deformation attribute of eyeglasses is distinct in all regions of the face, we introduce the face parsing model to segment the eyeglasses regions and attach extra regression loss to the regions to mitigate the negative affect of the global deformation.
- We conduct experiments on the MEGC2022 (generation track) datasets, and finally obtain the first place in expert evaluation.

2 METHODS

The framework of our whole model is shown in Fig.2, which is composed of four networks, which are background motion predictor E_{bg} , keypoint detector E_{kp} , dense motion network and inpainting network. The inputs of our model are source image I^S and driving image I^D sampled from the same video. The target of the training is to reconstruct the driving image with source image in an unsupervised way. The generation process can be divided into two parts, optical flow estimation and image inpainting. In the first process, the keypoint detector predict K group keypoints of Driving and source image (Noted as P^D and P^S respectively) and estimate the transformation with TPS method. Each group contains N keypoints. At the same time the background motion predictor estimate the affine transformation of the background. Then the dense motion network takes the K transformation and the affine transformation as the inputs to compute the optical flow and multi-stage occlusion maps. Finally, the inpainting network reconstructs driving images with optical flow and occlusion maps.

2.1 Motion Estimation

Face expression can be viewed as the deformation of the face. And from this perspective, face expression generation can be viewed as a transformation to deform the face from one shape to another. Thus, the most important part of the face expression transfer is the deformation estimation. Most previous methods estimate the deformation using linear methods, including zero order and first order method. However, these linear methods are actually not efficient enough to model the deformation of the face. Thus, inspired by [15] we adopted TPS method to estimate the deformation. Compared to linear methods, TPS is a flexible and powerful method. Given two sets of points, TPS can warp one set to the other with minimum distortion by using the TPS Transform T_{tps} :

$$\min \iint_{R^2} ((\frac{\partial^2 T_{tps}}{\partial x^2})^2 + 2(\frac{\partial^2 T_{tps}}{\partial x \partial y})^2 + (\frac{\partial^2 T_{tps}}{\partial y^2})^2) dx dy, \quad (1)$$

$$s.t. \quad T_{tps}(P_i^S) = P_i^D, i = 1, 2, ..., N$$

where the P_i^S and P_i^D are i -th corresponding keypoints of source and driving image. In our method, the detector detects K group keypoints and each group contains N keypoints. Therefore we can obtain K deformation with keypoints. The solving method

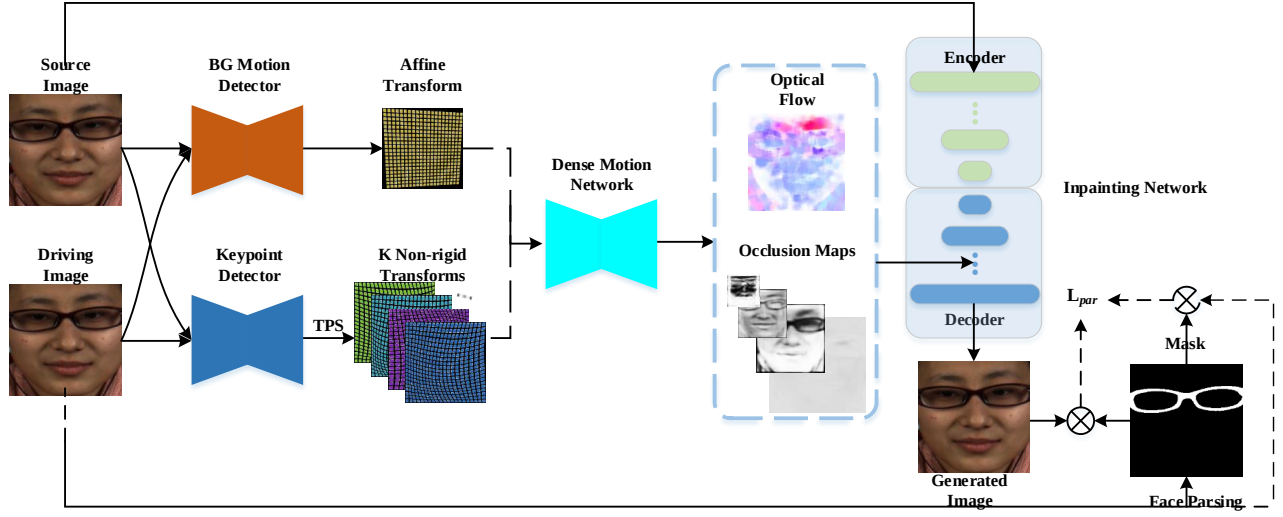


Figure 2: The framework of our solution.

the problem defined in Eq.1 has gone beyond the scope of our discussion, If you are interested in it, please refer to [1]. Here, we directly use the result to estimate the motion, which is shown as follows,

$$T_k(p) = A_k \begin{bmatrix} p \\ 1 \end{bmatrix} + \sum_{i=1}^N w_{ki} U(\|p_{ki}^D - p\|_2) \quad (2)$$

$$U(r) = r^2 \log(r^2) \quad (3)$$

where $p = [x, y]^T$. A_k and w_{ki} can be obtained by solving Eq.1, and U is a radial basis function.

The K group keypoints can produce K transformations while the background predictor generate one affine transformation. The $K + 1$ transformations are concatenated as the input of the dense motion network to produce optical flow of the faces and multi-stage occlusion maps. After that, the optical flow warp the source features to the targets and the occlusion maps help to attend to the missing region in feature warp. Both of them contribute to transfer the motion from the driving images to source images.

2.2 Face Parsing

Human faces can be divided into several regions (eyeglasses included). Among these regions, the eyeglasses region is a special one, as its deformation is rigid, while the deformation of others are non-rigid. Most methods process the image as a whole and tend to estimate the global optical flow, which affects the eyeglasses severely with non-rigid deformation. To mitigate this problem, we decide to introduce face parsing method to segment the eyeglasses regions and then apply additional regression loss to it.

Face parsing is in the scenario of the image segmentation task, which refers to segment the face into different facial parts. In our method, we adopt the deeplabv3[3] model trained with 14 classes and get the mask of the eyeglasses regions. The detail of the loss applied to the training is shown in the following section.

2.3 Losses

In the training phase, we make use of several different losses. The details of each loss function are shown as follows.

Similar to FOMM and some other methods in this field, we adopt reconstruction loss to ensure the content of driving and generated images are similar. To compute the reconstruction loss, we adopt a pretrained VGG19[14] net and compare the features of multi-resolution images are:

$$L_{rec} = \sum_j \sum_i |V_i(I_j^D) - V_i(I_j^{\hat{D}})| \quad (4)$$

where V_i means the i -th layer of the pretrained models, while I_j^D and $I_j^{\hat{D}}$ means the driving images and reconstructed images.

Also, to constrain the keypoint detector, we introduced the equivariance loss[12, 13],

$$L_{eq} = |T_{random}(E_{kp}(S)) - E_{kp}(T_{random}(S))| \quad (5)$$

where the T_{random} means a random TPS deformation.

Besides, we introduce the background loss to constrain and stabilize the training of the background motion predictor.

$$L_{bg} = \left| \begin{bmatrix} A_{sd} \\ 0, 0, 1 \end{bmatrix} \begin{bmatrix} A_{ds} \\ 0, 0, 1 \end{bmatrix} - I \right| \quad (6)$$

The $\begin{bmatrix} A_{sd} \\ 0, 0, 1 \end{bmatrix}$ means the affine transformation from source images to driving images, while $\begin{bmatrix} A_{ds} \\ 0, 0, 1 \end{bmatrix}$ means the transformation in the reverse direction. We expect the two transformation matrices to remain consistent. For inpainting network, we apply the warp loss,

$$L_{warp} = \sum_i |\hat{T}(E_i(I^S)) - E_i(I^D)| \quad (7)$$

where E_i is the i -th layer of the encoder in the inpainting network. The last loss function is the parsing Loss, which is utilized to ensure

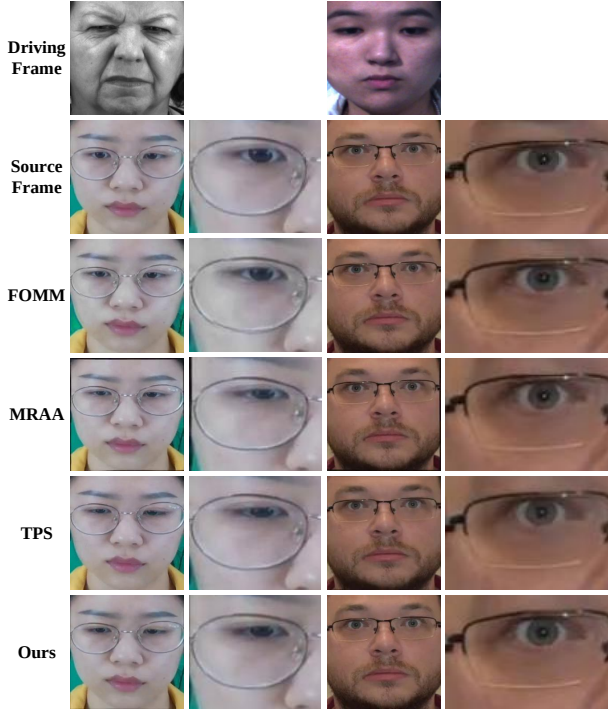


Figure 3: Results of several methods

the deformation of the eyeglasses regions is reasonable.

$$L_{par} = |I^{par} \times (I^D - \hat{I}^D)| \quad (8)$$

where I^{par} means the mask of the eyeglasses regions. In sum, the final loss can be written as,

$$L = \lambda_{rec}L_{rec} + \lambda_{eq}L_{eq} + \lambda_{bg}L_{bg} + \lambda_{warp}L_{warp} + \lambda_{par}L_{par} \quad (9)$$

3 EXPERIMENTS

3.1 Settings

Our proposed method is realized using the Pytorch framework and run on a platform with $4 \times$ V100S 32G GPUs. Our training epoch is set to 100 and the learning rate decays at the 50th and 90th epoch. the coefficient of the L_{par} is 50 while others are set to 10. The training takes about 30 hours. We conduct experiments on the three datasets provided by MEGC2022, i.e. CASMEII[7], SAMM[4] and SMIC-HS[8]. All the images are cropped before training according to the facial landmarks detected by dlib. All the images in the same sequence share the same bounding box in cropping. What's more, experiments are also conducted using MindSpore.

3.2 Comparison

We compare our method with three other methods: FOMM, MRAA and TPS. The results of three typical methods are shown in Fig.3. As the deformation of the micro-expression is too slight, we crop and enlarge the most informative parts of the face to make the visual comparison practicable. In Fig.3, we can observe that the quality of images generated with FOMM method and MRAA method is good but not so satisfactory. The MRAA method suffers from the missing pixels at the edge of the images, while the FOMM method cannot generate sharp edges of the eyeglasses. Moreover, the deformation

Table 1: Results of expert evaluation (normalized / Perfect_score)

Team	Coder 1	Coder 2	Coder 3	Total
36	0.61	0.56	0.44	1.61
46	0.73	0.55	0.58	1.86
63 (Ours)	0.74	0.58	0.66	1.98

Table 2: Results of expert evaluation (normalized / max_submission_perCoder)

Team	Coder 1	Coder 2	Coder 3	Total
36	0.83	0.97	0.66	2.46
46	1.00	0.95	0.88	2.82
63 (Ours)	1.00	1.00	1.00	3.00

of the generated faces using FOMM is not as obvious as other methods, which may attribute to ineffective representative ability of the first order approximation method.

TPS method performs better than FOMM and MRAA method in terms of amplitude of motion and quality of images, which verifies the effectiveness of the TPS motion estimation method. However, we find that the deformation of the glasses is not so natural with this method. In our proposed method, we modify the the TPS method by introducing the face parsing map to attend to the deformation of the eyeglasses regions. In this way, the unnatural deformation of the eyeglasses regions can be suppressed and better generation results can be obtained. In the last two rows of the Fig. 3, we can find that the undesired deformation is mitigated in our method, and the deformation of eyeglasses regions is more consistent to the deformation of the whole face compared with the TPS method.

Moreover, we also compare our method with the proposed methods of all other competitors in MEGC2022. After expert evaluation, our team (USTC-IAT-United) outperforms all other competitors and win the first prize, which verifies the effectiveness of our method (See Tab. 1, Tab.2).

4 CONCLUSION

In this paper, we propose a motion transfer network to transfer the micro-expression of the driving videos to the source images. The key of transfer is to model the deformation. As the intrinsic complexity of the face motion and different attributes of face regions, linear methods are hard to model it. Thus, we adopt TPS method to estimate the optical flow of the face motion. Moreover, we find that regions of eyeglasses show weird deformation in generation process. To mitigate this problem, we introduce face parsing method to apply extra constraints to the eyeglasses regions to ensure the reasonability of the deformation. We (Team: USTC-IAT-United) conduct several experiments on the provided datasets of the MEGC2022 and finally win the first prize in the expert evaluation.

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